

## STPM PHYSICS • SEMESTER 1 • CHAPTER 7

# Rotation of Rigid Bodies

## 7.1 Torque and Angular Acceleration

### DEFINITIONS & LAWS

- **Torque (moment of a force)** about an axis is the product of the force and the perpendicular distance from the axis to the line of action of the force.
- **Couple** is a pair of equal and opposite parallel forces whose lines of action do not coincide; it produces rotation without translation, and its torque is the product of one force and the perpendicular distance between the two forces.
- **Newton's second law of rotation:** the angular acceleration of a rigid body about a fixed axis is directly proportional to the resultant torque and inversely proportional to its moment of inertia.
- Torque is a vector (clockwise/anticlockwise sense must be stated); a force whose line of action passes through the axis has zero torque.
- **Angular acceleration**  $\alpha$  is the rate of change of angular velocity,  $\alpha = \frac{d\omega}{dt}$  ( $\text{rad s}^{-2}$ ).

### FORMULAE

| QUANTITY / LAW                             | FORMULA                                                                                                                  | SYMBOLS, UNITS & CONDITIONS                                                                                                                                                                                                                                                       |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Torque of a force</b>                   | $\tau = Fr \sin\theta = F_{\perp}r = Fd$                                                                                 | $\tau$ = torque (N m); $F$ = force (N); $r$ = distance from axis (m); $\theta$ = angle between $F$ and $r$ ; $F_{\perp}$ = component of $F$ perpendicular to $r$ ; $d = r \sin\theta$ = perpendicular distance (m)                                                                |
| <b>Couple</b>                              | $\tau = Fd$                                                                                                              | $\tau$ = torque of the couple (N m); $F$ = magnitude of each force (N); $d$ = perpendicular separation of the two forces (m)                                                                                                                                                      |
| <b>Torque and angular acceleration</b>     | $\tau = I\alpha$                                                                                                         | $\tau$ = torque (N m); $I$ = moment of inertia ( $\text{kg m}^2$ ); $\alpha$ = angular acceleration ( $\text{rad s}^{-2}$ )                                                                                                                                                       |
| <b>Linear-rotational links</b>             | $s = r\theta$ ; $v = r\omega$ ; $a_t = r\alpha$                                                                          | $s$ = arc length (m); $v$ = tangential speed ( $\text{m s}^{-1}$ ); $a_t$ = tangential acceleration ( $\text{m s}^{-2}$ ); $\theta$ = angle (rad); $\omega$ = angular velocity ( $\text{rad s}^{-1}$ ); $\alpha$ = angular acceleration ( $\text{rad s}^{-2}$ ); $r$ = radius (m) |
| <b>Equations of rotational motion</b>      | $\omega = \omega_0 + \alpha t$ ; $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$ ; $\omega^2 = \omega_0^2 + 2\alpha\theta$ | $\omega_0, \omega$ = initial and final angular velocities ( $\text{rad s}^{-1}$ ); $\theta$ = angular displacement (rad); $t$ = time (s); constant $\alpha$ ( $\text{rad s}^{-2}$ )                                                                                               |
| <b>Wheel or pulley with a hanging mass</b> | $mg - T = ma$ ; $Tr = I\alpha$ ; $a = r\alpha$                                                                           | $m$ = hanging mass (kg); $T$ = tension (N); $a$ = linear acceleration of mass ( $\text{m s}^{-2}$ ); $r$ = radius of wheel (m); $I$ = moment of inertia ( $\text{kg m}^2$ ); $\alpha$ = angular acceleration ( $\text{rad s}^{-2}$ )                                              |

## HOW TO ANSWER (MPM STYLE)

- Draw the force and the axis, mark the perpendicular distance  $d$ , then write  $\tau = F d$ ; state clockwise or anticlockwise.
- For  $\tau = I\alpha$  problems, find the net torque about the axis (sum of torques with sign), then  $\alpha = \frac{\tau}{I}$ ; add a linear equation for each hanging mass.
- Convert  $\text{cm} \rightarrow \text{m}$  and  $\text{rev min}^{-1} \rightarrow \text{rad s}^{-1}$  before substituting; give  $\tau$  in  $\text{N m}$  and  $\alpha$  in  $\text{rad s}^{-2}$ .
- For a couple, state that the resultant force is zero but the resultant torque is  $Fd$ .

## SECTION A TRAPS

**Trap:** Using the distance along the rod without  $\sin\theta$ ; the perpendicular distance is  $r \sin\theta$  (or use  $F \sin\theta$ ). If  $F$  is perpendicular to the rod,  $\sin\theta = 1$ .

**Trap:** Torque is not measured in joules for this subtopic; unit is  $\text{N m}$  (do not mistake it for energy).

**Trap:** In pulley problems, using  $T = mg$  for the string tension; the hanging mass accelerates, so  $T = m(g - a)$ .

**Trap:** Using  $F = ma$  for rotation without  $I$ : a body with larger  $I$  has smaller  $\alpha$  for the same torque.

## TRICKS

**Trick:** Limiting case: if the line of action passes through the pivot,  $d = 0$  and  $\tau = 0$ .

**Trick:** Ratio method: for the same  $\tau$ ,  $\alpha \propto \frac{1}{I}$ ; doubling the radius of gyration ( $I \propto r^2$ ) quarters  $\alpha$ .

## 7.2 Moment of Inertia

### DEFINITIONS & LAWS

- **Moment of inertia**  $I$  of a body about an axis is the sum of the products of the mass of each particle and the square of its perpendicular distance from the axis,  $I = \sum mr^2$ ; it is the measure of the body's resistance to angular acceleration (rotational inertia).
- Moment of inertia depends on the mass, on how the mass is distributed about the axis, and on the position of the axis; the same body has different  $I$  about different axes.
- Moment of inertia is the rotational analogue of mass:  $\tau = I\alpha$  corresponds to  $F = ma$ .
- A body with its mass distributed further from the axis has a larger  $I$  and so is harder to start or stop rotating.
- A flywheel has most of its mass at the rim to give a large  $I$ , so that it resists changes in angular speed (stores rotational kinetic energy).

## FORMULAE

| QUANTITY / LAW            | FORMULA                                              | SYMBOLS, UNITS & CONDITIONS                                                                                                                  |
|---------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Point mass                | $I = mr^2$                                           | $I$ = moment of inertia ( $\text{kg m}^2$ ); $m$ = mass (kg); $r$ = perpendicular distance from the axis (m)                                 |
| System of masses          | $I = \sum m_i r_i^2 = m_1 r_1^2 + m_2 r_2^2 + \dots$ | $I$ = moment of inertia ( $\text{kg m}^2$ ); $m_i$ = mass of the $i$ -th particle (kg); $r_i$ = its perpendicular distance from the axis (m) |
| Thin hoop or ring         | $I = MR^2$                                           | $M$ = mass (kg); $R$ = radius (m); about its axis                                                                                            |
| Solid disc or cylinder    | $I = \frac{1}{2}MR^2$                                | $M$ = mass (kg); $R$ = radius (m); about its axis                                                                                            |
| Solid sphere              | $I = \frac{2}{5}MR^2$                                | $M$ = mass (kg); $R$ = radius (m); about a diameter                                                                                          |
| Thin spherical shell      | $I = \frac{2}{3}MR^2$                                | $M$ = mass (kg); $R$ = radius (m)                                                                                                            |
| Thin rod about its centre | $I = \frac{1}{12}ML^2$                               | $M$ = mass (kg); $L$ = length (m); axis through the centre                                                                                   |
| Thin rod about one end    | $I = \frac{1}{3}ML^2$                                | $M$ = mass (kg); $L$ = length (m); axis through one end                                                                                      |
| Radius of gyration        | $I = Mk^2$                                           | $k$ = radius of gyration (m); $M$ = mass (kg); $I$ = moment of inertia ( $\text{kg m}^2$ )                                                   |

## HOW TO ANSWER (MPM STYLE)

- Identify the axis first; calculate the distance  $r$  of each mass from that axis (perpendicular distance), square it, multiply by the mass, and add.
- For a composite body, add the individual  $I$  values about the same axis; show every term with units  $\text{kg m}^2$ .
- For comparing bodies of the same mass and radius, give the order by mass distribution: hoop > disc > solid sphere ( $\frac{I}{MR^2} = 1, 0.5, 0.4$ ).
- Link to  $\tau = I\alpha$  when asked for the angular acceleration of the body.

## SECTION A TRAPS

**Trap:** The distance in  $\sum mr^2$  is measured from the axis, not from the centre of the body or from the other mass; a mass lying on the axis contributes zero.

**Trap:** Doubling the distance of a mass from the axis multiplies its contribution by 4 ( $r^2$ ), not by 2.

**Trap:** Using the formula for the wrong axis: rod about the centre is  $\frac{ML^2}{12}$ , about the end is  $\frac{ML^2}{3}$  (four times larger); disc about its own axis is  $\frac{1}{2}MR^2$ .

**Trap:** Moment of inertia of a rigid body is not a fixed property of the body; it is different about different axes and changes if the mass distribution is changed.

## TRICKS

**Trick:** Keep the order  $I(\text{hoop}) > I(\text{disc}) > I(\text{solid sphere})$  for the same  $M$  and  $R$ , since less mass at the rim lowers  $I$ .

**Trick:** Unit check:  $I$  has unit  $\text{kg m}^2$ ; a result in  $\text{kg m}^{-2}$  or  $\text{kg m}$  indicates a missing or extra power of  $r$ .

## 7.3 Angular Momentum and Rotational Kinetic Energy

### DEFINITIONS & LAWS

- Angular momentum**  $L$  of a rigid body about an axis is the product of its moment of inertia and its angular velocity,  $L = I\omega$  ( $\text{kg m}^2 \text{s}^{-1}$ ); for a point mass  $L = mvr$ .
- Principle of conservation of angular momentum:** the total angular momentum of a system remains constant if no resultant external torque acts on the system.
- Rotational kinetic energy** is the energy a rotating body possesses because of its angular velocity,  $\frac{1}{2}I\omega^2$ .
- Newton's second law in terms of angular momentum: the resultant torque is the rate of change of angular momentum,  $\tau = \frac{dL}{dt}$ .
- When the moment of inertia of a system decreases (e.g. a skater draws in the arms),  $\omega$  increases so that  $L$  is constant; the rotational kinetic energy increases because work is done by the skater (internal forces).
- In a collision or coupling of rotating bodies about the same axis (inelastic), angular momentum is conserved but kinetic energy is lost.

### FORMULAE

| QUANTITY / LAW                   | FORMULA                                                 | SYMBOLS, UNITS & CONDITIONS                                                                                                                                                                                  |
|----------------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Angular momentum                 | $L = I\omega = mvr$                                     | $L$ = angular momentum ( $\text{kg m}^2 \text{s}^{-1}$ ); $I$ = moment of inertia ( $\text{kg m}^2$ ); $\omega$ = angular velocity ( $\text{rad s}^{-1}$ ); $mvr$ applies to a point mass in circular motion |
| Torque and angular momentum      | $\tau = \frac{dL}{dt} = I \frac{d\omega}{dt} = I\alpha$ | $\tau$ = torque ( $\text{N m}$ ); $\alpha$ = angular acceleration ( $\text{rad s}^{-2}$ ); $I$ constant                                                                                                      |
| Conservation of angular momentum | $I_1\omega_1 = I_2\omega_2$                             | $I$ = moment of inertia ( $\text{kg m}^2$ ); $\omega$ = angular velocity ( $\text{rad s}^{-1}$ ); no external torque                                                                                         |
| Coupling of two rotating bodies  | $I_1\omega_1 + I_2\omega_2 = (I_1 + I_2)\omega$         | $I_1, I_2$ = moments of inertia ( $\text{kg m}^2$ ); $\omega_1, \omega_2$ = initial angular velocities ( $\text{rad s}^{-1}$ ); $\omega$ = common final angular velocity ( $\text{rad s}^{-1}$ )             |

| QUANTITY / LAW                                | FORMULA                                                      | SYMBOLS, UNITS & CONDITIONS                                                                                                                                                                            |
|-----------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Rotational kinetic energy</b>              | $K = \frac{1}{2}I\omega^2 = \frac{L^2}{2I}$                  | K = rotational kinetic energy (J); I = moment of inertia ( $\text{kg m}^2$ ); $\omega$ = angular velocity ( $\text{rad s}^{-1}$ ); L = angular momentum ( $\text{kg m}^2 \text{s}^{-1}$ )              |
| <b>Rolling without slipping</b>               | $v = r\omega$ ; $K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ | v = speed of centre ( $\text{m s}^{-1}$ ); r = radius (m); $\omega$ = angular velocity ( $\text{rad s}^{-1}$ ); K = total kinetic energy (J); m = mass (kg); I = moment of inertia ( $\text{kg m}^2$ ) |
| <b>Rolling down an incline</b>                | $mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$               | h = height of incline (m); starts from rest; m = mass (kg); v = speed at the bottom ( $\text{m s}^{-1}$ ); $\omega$ = angular velocity ( $\text{rad s}^{-1}$ )                                         |
| <b>Solid cylinder rolling down an incline</b> | $v = \sqrt{\frac{4gh}{3}}$                                   | v = speed at the bottom ( $\text{m s}^{-1}$ ); h = height (m); g = gravitational acceleration ( $\text{m s}^{-2}$ ); from rest                                                                         |
| <b>Work done by a torque</b>                  | $W = \tau\theta$                                             | W = work done (J); $\tau$ = torque (N m); $\theta$ = angle turned (rad)                                                                                                                                |
| <b>Power of a torque</b>                      | $P = \tau\omega$                                             | P = power (W); $\tau$ = torque (N m); $\omega$ = angular velocity ( $\text{rad s}^{-1}$ )                                                                                                              |

## HOW TO ANSWER (MPM STYLE)

- State the principle of conservation of angular momentum with the condition "no resultant external torque" before using  $I_1\omega_1 = I_2\omega_2$ .
- Find I for each stage ( $I = \Sigma mr^2$ ), substitute with  $\omega$  in  $\text{rad s}^{-1}$ , and give the final  $\omega$  with unit  $\text{rad s}^{-1}$  (convert to  $\text{rev min}^{-1}$  only if asked).
- When asked for the change in kinetic energy, calculate  $\frac{1}{2}I\omega^2$  before and after and state the source (work done by the person/internal forces) or the loss (inelastic).
- For rolling, include both translational and rotational kinetic energy; state  $v = r\omega$  as the no-slip condition.

## SECTION A TRAPS

**Trap:** Only L is conserved when I changes;  $\frac{1}{2}I\omega^2$  is not conserved. If I decreases by half,  $\omega$  doubles and K doubles ( $K = \frac{L^2}{2I}$ ).

**Trap:** Angular momentum is conserved only with zero resultant external torque; do not claim it for a body acted on by a net torque.

**Trap:** Rolling: total kinetic energy is greater than  $\frac{1}{2}mv^2$ ; a body with a smaller  $\frac{I}{MR^2}$  (solid sphere) reaches the bottom of the incline first and has the larger v, independent of m and R.

**Trap:** Forgetting to convert  $\text{rev min}^{-1}$  to  $\text{rad s}^{-1}$  before using  $\frac{1}{2}I\omega^2$ ; this gives an error of a factor  $(\frac{60}{2\pi})^2$ .

## TRICKS

**Trick:** Skater:  $I_2 = \frac{I_1}{n}$  gives  $\omega_2 = n\omega_1$  and  $K_2 = nK_1$ ; use the ratio directly without the numbers.

**Trick:** Unit check:  $\frac{1}{2}I\omega^2$  gives  $\text{kg m}^2 \text{s}^{-2} = \text{J}$ ; if the unit is not J,  $\omega$  has not been in  $\text{rad s}^{-1}$ .